

Sprint Manifolds and Cobordism Dynamics: A Transition Function Framework for Varying Constants and Gauge Group Selection in Cosmological Cycle Models

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Abstract

We propose a framework in which the fundamental constants c , $\hbar$, and G are outputs of a sprint transition function T operating on the compactification geometry inherited across cosmological cycle boundaries, rather than free parameters. The framework advances nine principal claims. (RT-1) The constants c , $\hbar$, G vary across sprint cycles as outputs of T . (RT-2) The gauge group of each sprint is determined by compactification geometry via sprint transition dynamics. (RT-3) The global structure of all sprint cycles is the Meta-Manifold $\mathcal{M}$, a cobordism category of 11-dimensional M-Theory patches connected by 12-dimensional transition manifolds. (RT-4) The initial boundary datum Ω_0 of $\mathcal{M}$ constitutes the first uncaused boundary condition; its nature is an open question. (RT-5) The sprint uncertainty $\hbar_n/2$ is the irreducible non-closure condition on dynamics within each sprint; $\mathcal{M}$ is the sub-category of cobordism sequences maintaining $\hbar_n > 0$ at every transition. (RT-6) Effective physics within a sprint arises from the Plebanski mechanism applied to the sprint's gauge group via spontaneous symmetry breaking, following Lisi, Smolin, and Speziale (2010). (RT-7) The three fermion generations correspond to the $\mathbb{Z}_3$ triality of $D_4 = \mathfrak{so}(8)$. (RT-8) Our sprint's gauge history descends from E_6 through the GUT chain to the Standard Model. (RT-9) The transition function T is specified as a two-stage chain (conformal rescaling of compact boundary data followed by Plebanski SSB), making RT-1 through RT-3 falsifiable in principle. The Distler–Garibaldi chirality theorem is addressed through three independent arguments, including a quaternionic unity construction via the $E_{8(-24)}$ real form. The framework integrates M-Theory, Penrose's Conformal Cyclic Cosmology, the Lisi–Smolin–Speziale Plebanski unification, the Freudenthal–Tits Magic Square, and G_2 holonomy mathematics into a single coherent structure. Every substantive claim carries an explicit epistemic status label.

Version note: This document carries version number V0.3. It will not reach V1.0 without experimental confirmation at the Planck scale, which no current technology can provide. The versioning is not modesty. It is precision.

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Claim status key: Every substantive claim carries one of four labels. [ESTABLISHED] Proven, peer-reviewed, undisputed. [SPECULATIVE] Proposed in the physics literature, plausible, not yet confirmed. [RENSHAW] Novel claim of this framework. [OPEN] Question raised but not answered.

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1 Foundation: The Priority of Geometry

The standard formulation of fundamental physics treats the constants of nature as given — measured parameters whose values are inputs to the equations rather than outputs of a deeper structure. Renshaw Theory inverts this: the geometric substrate of reality determines the constants, not the reverse. The constants are outputs of a prior geometric instantiation.

Applied to physics: the values of c , $\hbar$, and G are not universal constants across cosmological cycles. They are functions of the compactification geometry inherited at each cycle boundary. The geometry is primary. The constants are derived.

The initial boundary datum Ω_0 — whatever determined the compactification geometry of the first sprint cycle — is an open question (RT-4). This framework identifies it as the first boundary condition of $\mathcal{M}$ and acknowledges that its nature lies outside the scope of physical falsification. The mathematical structure of the framework does not depend on the answer.

2 The Speed of Light as Sprint Geometry

2.1 What c Actually Is

[ESTABLISHED]

$c = 299,792,458$ m/s is not merely the speed at which light travels. Since 1983 it defines the metre. More fundamentally, c is a structural constant of spacetime itself. Maxwell's equations give $c = 1/\sqrt{\varepsilon_0\mu_0}$, where ε_0 and μ_0 are properties of the vacuum manifold, not of any substance within it. Einstein's Special Relativity established the invariance of c as a consequence of the geometric structure of Minkowski spacetime:

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2 \quad (1)$$

c is not a speed limit imposed on particles. It is the speed at which zero-mass excitations of the manifold propagate — the propagation speed of the geometry itself.

2.2 The Planck System

[ESTABLISHED]

In Planck units ($c = \hbar = G = 1$), the natural scales are:

$$l_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35} \text{ m} \quad (2)$$

$$t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44} \text{ s} \quad (3)$$

$$m_P = \sqrt{\hbar c/G} \approx 2.176 \times 10^{-8} \text{ kg} \quad (4)$$

Below the Planck scale, the smooth manifold description of spacetime breaks down. l_P , t_P , and m_P are all functions of c , $\hbar$, and G simultaneously; they are not independent — three expressions of the same geometric structure. In this framework, this interdependence is primary: the Planck scale is the resolution limit of the current sprint.

2.3 RT-1: c as Sprint Output

[RENSHAW] Coherent with M-Theory landscape. Not testable within current technology.

In M-Theory, the effective 4-dimensional physics depends on how the extra 7 dimensions are compactified. Different compactification geometries produce different effective values of c , $\hbar$, and G . If each cosmic sprint cycle is a topological patch on a larger manifold, the compactification geometry of each patch is determined by the boundary conditions inherited from the previous patch.

RT-1: c , $\hbar$, and G are not universal constants across sprint cycles. They are outputs of the sprint transition function T applied to the compactification geometry $\mathcal{G}_n$ inherited from sprint $n - 1$:

$$c_n = f(\mathcal{G}_n), \quad \mathcal{G}_n = T(\mathcal{G}_{n-1}, \delta_{n-1}) \quad (5)$$

where δ_{n-1} encodes the time-dilation history of sprint $n - 1$. In sprint 0, $\mathcal{G}_0$ is the initial boundary datum Ω_0 of the Meta-Manifold (RT-4).

2.4 c and the ADM Formalism

The ADM formalism [1] provides the direct technical connection. The lapse function N and shift vector N^i of the spacetime foliation encode the propagation structure: c emerges as the ratio between the lapse function and the induced metric on the spatial hypersurface. In a sprint manifold, this foliation is determined by the compactification geometry $\mathcal{G}_n$, making c a geometrically determined output rather than a free parameter.

3 Sprint Manifolds, Superspace, and the Meta-Manifold

3.1 Bosonic and Fermionic Structure in Superspace

[ESTABLISHED]

Penrose and Rindler, in *Spinors and Space-Time* [5, 6], develop the formalism of supernumbers. A supernumber z decomposes as $z = z_B + z_S$, where z_B is the bosonic (commuting) component and z_S is the fermionic (anticommuting, Grassmann) component. In superspace, the coordinates are:

$$(x^\mu, \theta^\alpha, \bar{\theta}^{\dot{\alpha}}) \quad (6)$$

where x^μ are spacetime coordinates (bosonic) and $\theta^\alpha, \bar{\theta}^{\dot{\alpha}}$ are Grassmann-valued fermionic coordinates. The distinction is not arbitrary: bosons commute, fermions anticommute.

In 11-dimensional supergravity — the low-energy limit of M-Theory — spinors are 32-component Majorana spinors. The sprint superspace has 11 bosonic dimensions and 32 fermionic dimensions. The counting $2^{\lceil 11/2 \rceil} = 32$ is itself a consequence of the Clifford algebra dimension formula: the number of fermionic directions is determined by the bosonic dimensionality, not chosen independently. The 32 fermionic directions of sprint superspace are a mathematical necessity, not a free parameter.

3.2 Sprint Manifold Taxonomy

[RENSHAW] Cobordism mathematics is established; its application to cosmic sprint cycles is novel.

Sprint Manifold SM_n

The 11-dimensional M-Theory manifold of the n -th cosmic sprint cycle. Locally $SM_n \approx M^{1,3} \times \Sigma^7$, where Σ^7 is the 7-dimensional compact internal space.

Sprint Boundary ∂SM_n

The conformal boundary of SM_n , analogous to Penrose’s cross-sections in Conformal Cyclic Cosmology [7]. The boundary data encodes the deliverables of sprint n : the constants, the compactification geometry, the time-dilation history.

Sprint Cobordism $SC_{n,n+1}$

The 12-dimensional transition manifold connecting ∂SM_n to ∂SM_{n+1} . Encodes the inheritance rules: how $\mathcal{G}_{n+1}$, c_{n+1} , and $\hbar_{n+1}$ are determined by the boundary data of sprint n .

Meta-Manifold $\mathcal{M}$

The colimit of the cobordism category $\{SM_n, SC_{n,n+1}\}$. The global structure of all sprint cycles. $\mathcal{M}$ is a higher categorical object, not itself a Riemannian manifold.

The McNamara–Vafa Cobordism Conjecture [4] provides independent convergence on this structural claim from within the Swampland program. McNamara and Vafa argued that the space of consistent quantum gravity theories is constrained by cobordism: any two consistent vacua connected by a cobordism must be related by a physical transition. This is precisely the structure RT-3 proposes at the cosmological level. The sprint cobordism category is a specific instantiation of the cobordism constraint on the landscape of consistent physics.

3.3 RT-2 Through RT-5 — Formal Statement

RT-2 [RENSHAW]: The gauge group of sprint n is determined by the compactification geometry $\mathcal{G}_n$, which is inherited from sprint $n - 1$ via the cobordism SC . It is not freely chosen.

RT-3 [RENSHAW]: The global structure of all sprint cycles is the Meta-Manifold $\mathcal{M}$, the colimit of the sprint cobordism category: a category whose objects are 11-dimensional M-Theory sprint manifolds (SM_n) and whose morphisms are 12-dimensional transition cobordisms ($SC_{n,n+1}$).

RT-4 [OPEN]: The initial boundary datum Ω_0 is the uncaused first boundary condition that determines $\mathcal{G}_0$ of sprint 0. The nature of Ω_0 lies outside the scope of this framework’s falsifiable claims. The mathematical structure of $\mathcal{M}$ requires Ω_0 to exist but does not determine its origin.

RT-4 Extended (Attractor Structure) [RENSHAW]: In the dynamical-systems structure of T , Ω_0 may function not only as the initial condition but also as a terminal attractor — the boundary condition the dynamics converge toward. Whether this is the case is an open question with falsifiable consequences (see Section 7).

RT-5 [RENSHAW]: The sprint uncertainty $\hbar_n/2$ is the minimum quantum of non-closure in sprint n . $\mathcal{M}$ is the sub-category of cobordism sequences that maintain $\hbar_n > 0$ at every transition. The $\hbar_n > 0$ condition ensures non-trivial quantum dynamics within each sprint; sprints with $\hbar_n = 0$ would be classically deterministic and structurally degenerate.

4 Spinors — Algebra, Geometry, and Matter Structure

4.1 The Belt Trick: What Spinors Are

[ESTABLISHED]

A vector returns to its original value after a 360° rotation. A spinor acquires a phase of -1 after 360° and returns to its original value only after 720° . This is physically demonstrated by the Dirac belt trick. The topology is exact: $\pi_1(SO(3)) = \mathbb{Z}_2$. There are two topologically distinct paths in $SO(3)$. $SU(2)$, the double cover, resolves them. Half of all matter — all fermions: electrons, quarks, neutrinos — is described by spinors.

4.2 The $SU(2)$ Algebra and Dirac Equation

[ESTABLISHED]

A two-component Weyl spinor ξ^α transforms under $SU(2)$ as $\xi^\alpha \rightarrow M^\alpha_\beta \xi^\beta$. The $SU(2)$ generators are the Pauli matrices σ_i satisfying $[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$. For a rotation by 2π : $M = -\mathbf{1}$, confirming 720° periodicity.

The Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ encodes the spacetime metric into the Clifford algebra of the gamma matrices $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$. The fermionic structure of matter is its spinor structure.

4.3 The 32-Component Majorana Spinor in 11D

[ESTABLISHED]

In 11-dimensional M-Theory, the minimal spinor representation has $2^{\lfloor 11/2 \rfloor} = 2^5 = 32$ components. This is a consequence of the Clifford algebra dimension formula, not a choice. The 32 fermionic directions of M-theory superspace are determined by the 11 bosonic dimensions via mathematical necessity. The fermionic structure is not independently specified; it is entailed by the bosonic dimensionality.

5 E_8 , the Octonion Tower, and the Chirality Response

5.1 The Hurwitz Theorem and the Octonion Tower

[ESTABLISHED]

Hurwitz's theorem (1898): exactly four normed division algebras exist — $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ (quaternions), $\mathbb{O}$ (octonions). Each step costs one algebraic property: order, commutativity, associativity. The sedenions (dimension 16) lose division entirely. The tower terminates at $\mathbb{O}$.

Every exceptional Lie algebra arises from octonions via the Freudenthal–Tits Magic Square [8]: $G_2 = \text{Aut}(\mathbb{O})$, F_4 from $H_3(\mathbb{O})$, and E_6 , E_7 , E_8 from combinations. $E_8 = \mathcal{M}(\mathbb{O}, \mathbb{O})$: both inputs octonionic. E_8 is terminal because $\mathbb{O}$ is terminal.

5.2 The Plebanski Mechanism and Coleman–Mandula

[SPECULATIVE], building on established results.

Lisi, Smolin, and Speziale [3] extended the Plebanski action for GR to a gauge field valued in a Lie algebra that spontaneously breaks to gravity + Yang–Mills + Higgs. The

Coleman–Mandula loophole is genuine: the theorem forbids mixing of global spacetime and internal symmetries. It says nothing about local gauge theories that spontaneously break into separate sectors. All low-energy coupling constants emerge as functions of a single parameter λ and the Higgs vacuum expectation value.

RT-6 [RENSHAW]: Effective physics within a sprint arises from spontaneous symmetry breaking of the sprint’s gauge group via the Plebanski mechanism. All Standard Model parameters are functions of a single coupling constant λ_n determined by the sprint transition.

5.3 The Chirality Response

[RENSHAW]

The Distler–Garibaldi theorem [2] proves that no real form of E_8 admits a chiral embedding of the Standard Model fermion spectrum. The response proceeds on three independent arguments:

Argument 1: Avoidance by construction. The framework does not embed the Standard Model fermion spectrum directly in E_8 . Instead, the Plebanski mechanism spontaneously breaks the gauge symmetry after the sprint transition, and the effective chiral spectrum emerges from the pattern of symmetry breaking, not from a direct group-theoretic embedding.

Argument 2: $E_{8(-24)}$ preference. The split real form $E_{8(8)}$ is the natural choice in string theory. But the quaternionic real form $E_{8(-24)}$ has maximal compact subgroup $E_7 \times SU(2)$, which provides a natural bosonic/fermionic decomposition in the sprint superspace: E_7 governs the 7-dimensional compact space (Σ^7) and $SU(2)$ governs the spinor structure.

Argument 3: Quaternionic unity. $G_2 = \text{Aut}(\mathbb{O})$ and $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}$ (as real vector spaces). The G_2 holonomy preference in M-Theory compactifications on 7-manifolds connects directly to the $E_{8(-24)}$ preference via the quaternionic structure of the octonions. The isomorphism chain runs: $G_2 \hookrightarrow SO(7) \hookrightarrow SO(8)$ (with triality), and the G_2 -invariant subspace of $\mathbb{O}$ is the imaginary quaternions $\text{Im}(\mathbb{H})$. This means the holonomy group that selects the compact geometry and the real form that governs the gauge structure are both controlled by the same quaternionic substructure of the octonions.

6 The Transition Function T

This section specifies the transition function T that governs the passage from sprint n to sprint $n + 1$. This is where the framework moves from structure to falsifiable proposal.

6.1 The Two-Stage Chain

T operates in two stages:

Stage 1: Conformal rescaling. The boundary data ∂SM_n is conformally rescaled to produce the initial conditions for SM_{n+1} , following the Penrose CCC prescription [7] but applied to the full 11-dimensional sprint rather than the 4-dimensional effective spacetime.

Stage 2: Plebanski SSB. The rescaled geometry determines a new compactification $\mathcal{G}_{n+1}$, from which the gauge group of sprint $n + 1$ emerges via the Plebanski mechanism. The coupling constant λ_{n+1} is a function of $\mathcal{G}_{n+1}$, and all Standard Model parameters of sprint $n + 1$ follow.

6.2 Five Constraints on Φ

The transition field Φ (the data carried across the sprint cobordism) must satisfy:

1. **M-Theory admissibility:** Φ must produce a geometry consistent with 11-dimensional supergravity.
2. **G_2 holonomy:** The compact 7-manifold Σ_{n+1}^7 must admit G_2 holonomy for $\mathcal{N} = 1$ supersymmetry in the effective 4D theory.
3. **Cobordism consistency:** The transition must satisfy the cobordism constraints of the McNamara–Vafa conjecture [4].
4. **Generation count:** The resulting spectrum must contain exactly three fermion generations (via D_4 triality, RT-7).
5. **$\hbar$ preservation:** $\hbar_{n+1} > 0$ (the non-closure constraint, RT-5).

6.3 Falsification Criterion

If the Standard Model parameters are demonstrably independent with no algebraic relations derivable from a single coupling λ , then RT-6 and the transition function specification are falsified. The specific prediction is not merely that some relations among SM parameters exist — approximate relations such as the Koide formula for charged lepton masses, the Cabibbo angle relations, and the approximate b – τ unification in SUSY GUTs are already known — but that *all* relations are derivable from a single λ . That is a sharper claim than “some correlations exist,” and it deserves to be stated sharply.

7 Dynamical Systems Structure

7.1 The Self-Referential Form of T

The sprint geometry determines the coupling constants which determine the conformal factor which determines the next sprint geometry. This is a genuine fixed-point problem:

$$\mathcal{G}^* = T(\mathcal{G}^*, \delta^*) \tag{7}$$

The landscape problem is thereby recast: instead of selecting among $\sim 10^{500}$ vacua by anthropic reasoning, the question becomes which attractors of T are dynamically accessible from Ω_0 . This is attractor enumeration, not landscape selection.

7.2 The ADE-Attractor Conjecture

Conjecture (ADE-Attractor) [RENSHAW]/[OPEN]: There exists a correspondence between the fixed points of the sprint transition map T and the ADE classification of G_2 singularities. If confirmed, the attractor structure of the sprint transition would naturally select among gauge groups — because the ADE classification appears in both G_2 singularities (determining gauge content via M-Theory on singular G_2 manifolds) and in the Dynkin diagrams of the possible gauge groups. The fixed points of T would then correspond to ADE singularity types, meaning the dynamics of sprint evolution select the physics rather than the physics being selected by fiat.

This conjecture is named explicitly to make it citable and to give the mathematical physics community a specific target.

7.3 RT-5 Formalised: Non-Closure and Attractor Selection

The non-closure condition $\hbar_n > 0$ has a structural role beyond quantum mechanics: it ensures that the dynamics within each sprint are non-deterministic in the classical sense, preventing degenerate fixed points of T from collapsing the cobordism structure. The distinction between “random” and “non-closed” is essential: $\hbar > 0$ is a necessary condition for non-trivial quantum dynamics. Whether it is also a sufficient condition for any deeper notion of indeterminacy is a question this framework raises but does not answer.

7.4 RT-7: Triality and Three Generations

[SPECULATIVE]

$SO(8)$ is the only simple Lie group whose Dynkin diagram has $\mathbb{Z}_3$ outer automorphism symmetry (triality). This permutes three 8-dimensional representations: $\mathbf{8}_v$ (vector), $\mathbf{8}_s$ (left spinor), $\mathbf{8}_c$ (right spinor). The Standard Model has three generations of fermions, each consisting of 8 species (6 quarks + 2 leptons). The correspondence in count (3) and dimension (8) is structurally compelling.

RT-7 [SPECULATIVE]: The three fermion generations correspond to the $\mathbb{Z}_3$ triality of $D_4 = \mathfrak{so}(8)$. Three is not a free parameter. It is the triality number of the octonions, which terminate the Hurwitz tower at dimension 8.

7.5 RT-8: The GUT Ladder

[SPECULATIVE]

The observed gauge group $SU(3) \times SU(2) \times U(1)$ descends from the heterotic visible-sector group E_8 through the GUT chain:

$$E_8 \supset E_7 \supset E_6 \rightarrow SO(10) \rightarrow SU(5) \rightarrow SU(3) \times SU(2) \times U(1) \quad (8)$$

The entry point E_6 is the natural heterotic visible sector in Calabi–Yau compactification. The breaking pattern is constrained by anomaly cancellation, proton decay limits, and the observed spectrum.

RT-8 [SPECULATIVE]: Our sprint’s gauge history descends from E_6 (the natural heterotic visible sector) through the GUT chain to the Standard Model. The breaking pattern is determined by $\mathcal{G}_n$ and T , not freely chosen.

8 Complete Claims Summary

Claim	Status	Statement	Falsifiable?
RT-1	[RENSHAW]	$c, \hbar, G$ are sprint outputs, not constants	In principle
RT-2	[RENSHAW]	Gauge group determined by $\mathcal{G}_n$	In principle
RT-3	[RENSHAW]	Meta-Manifold $\mathcal{M}$ as cobordism category	In principle
RT-4	[OPEN]	Initial boundary datum Ω_0 as first condition	No
RT-5	[RENSHAW]	$\hbar > 0$ as non-closure constraint	In principle
RT-6	[SPECULATIVE]	Plebanski mechanism for SM parameters	Yes
RT-7	[SPECULATIVE]	Triality $\Leftrightarrow$ three generations	In principle
RT-8	[SPECULATIVE]	E_6 GUT descent	In principle
RT-9	[RENSHAW]	Transition function T specification	Yes

Table 1: Complete claims registry, V0.3.

9 Scope: What This Framework Claims and Does Not Claim

What this framework claims:

- That the constants of nature are outputs of a transition function operating on cobordism structure.
- That the non-closure constraint ($\hbar > 0$) has structural significance for the dynamics of T .
- That the transition function specification makes the framework falsifiable in principle.
- That the ADE-Attractor Conjecture provides a specific, testable connection between sprint dynamics and gauge group selection.

What this framework does not claim:

- Experimental confirmation at the Planck scale.
- A unique derivation of the Standard Model parameters from first principles (this remains the programme, not the achievement).
- That RT-4 is physically testable. The nature of Ω_0 is an open question and clearly marked as such.
- That this framework is complete. V0.3 means V0.3.

10 Research Roadmap: Toward V0.4

1. **Foreground McNamara–Vafa convergence:** Develop the connection between the sprint cobordism category and the Swampland program’s cobordism constraints as independent convergence on RT-3.
2. **Expand the quaternionic unity argument:** The isomorphism chain $G_2 \hookrightarrow SO(7) \hookrightarrow SO(8)$ and the connection between G_2 holonomy preference and $E_{8(-24)}$ preference via $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}$ deserves fuller representation-theoretic treatment.

3. **Sharpen RT-6 falsification:** Tighten the falsification criterion with explicit reference to known approximate algebraic relations among Standard Model parameters (Koide formula, Cabibbo angle relations, b - τ unification).
4. **Develop the ADE-Attractor Conjecture:** Establish whether the fixed-point structure of T admits an ADE classification by examining specific G_2 compactification examples.
5. **ADM formalism development:** Full subsection showing how c emerges as a geometric ratio from the sprint manifold's foliation via the lapse function and shift vector.
6. **Clifford algebra entailment:** Emphasise that the $2^{\lfloor 11/2 \rfloor} = 32$ fermionic directions are a mathematical entailment of the bosonic dimensionality, not a design choice.

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The complete framework, including philosophical and metaphysical extensions beyond the scope of this paper, is available at V0.3 from the authors. The physics presented here stands independently of those extensions.

The authors made use of Claude (Anthropic, claude.ai), a large language model AI assistant, as a software tool during the preparation of this work. Per current arXiv and publisher policy, AI language models do not meet the criteria for authorship and are not listed as authors. All intellectual content, research decisions, and responsibility for the work remain solely with the human authors.

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